MATH 152 **QUIZ 5** NAME _____

Mrs. Bonny Tighe 25 points

7.8,8.1,8.2 TA_____Wed. 3/13/2024

1. The area under a curve can be found if the improper definite integral is _____

2. Describe the two types of improper integrals.

3. State the Comparison Test for Improper Integrals

4. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

a) $\int_2^5 \frac{1}{x^2 - 4}\, dx$

b) $\int_0^\infty \frac{1}{1 + x^2} dx$

5. Use the comparison test to determine whether the integrals diverge or converge.

$$\int_{2}^{\infty} \frac{3x^2}{\sqrt{x^5 - 4x - 3}}\,dx$$

6. Find the length of the arc of the parabola $y = \frac{x^3}{6} + \frac{1}{2x}$, $\frac{1}{2} \le x \le 1$

$$L = \int_{a}^{b} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$$

7. Set up an integral, **but do not solve**, for the area of the surface obtained by rotating the curve $y = \ln x, \quad 1 \le x \le 3$ about a) the x-axis and b) the y-axis

8. The given curve is rotated about the x-axis. Find the area of the resulting surface.

$y = \sqrt{1 + e^x}, \quad 0 \le x \le 1$ $\qquad S = 2\pi \int_a^b radius\,(arclength)$

MATH 152 **QUIZ 5A** NAME _____

Mrs. Bonny Tighe 25 points

7.8,8.1,8.2 TA_____Wed. 3/13/2024

1. State the Comparison Test for Improper Integrals

2. The area under a curve can be found if the improper definite integral is _____

3. Describe the two types of improper integrals.

4. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

a) $\int_1^2 \frac{1}{x\sqrt{x^2 - 1}}\, dx$

b) $\int_0^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx$

5. Use the comparison test to determine whether the integrals diverge or converge.

$$\int_2^{\infty} \frac{x^3}{\sqrt{x^7 - 4x - 3}}\,dx$$

6. Find the length of the arc $x = \frac{y^4}{8} + \frac{1}{4y^2}$ $1 \le y \le 2$ $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$

7. Set up an integral, **but do not solve**, for the area of the surface obtained by rotating the curve $y = e^x$, $1 \le x \le 3$ about a) the x-axis and b) the y-axis

8. The given curve is rotated about the x-axis. Find the area of the resulting surface.

$y = \sqrt{1 + e^x}, \quad 0 \le x \le 1 \qquad S = 2\pi \int_a^b radius(arclength))$